TSLA Tesla, Inc. — 个股深度研究 (专业中文版) v20261017-0755

> 数据截至 2026-10-17 07:55 SGT｜价格 $354.72｜市值 1,401.0 bn USD｜财报货币 USD｜非投资建议。事实=CONFIRMED/DATA；推断=INFERENCE；假设=ASSUMPTION/HYPOTHESIS。

1. 投资论点 Investment Thesis (HYPOTHESIS)

汽车主业利润率受价格战压缩，估值主要来自 Robotaxi/FSD 与 Optimus 期权以及储能高增长；股价对自动驾驶里程碑的叙事极度敏感(反身性强)。

- **多头驱动**: Unsupervised FSD scales across cities; Robotaxi unit economics proven; storage margins stay high｜无监督 FSD 多城扩张；Robotaxi 单位经济验证；储能高毛利
- **空头驱动**: Auto volumes/margins keep falling; robotaxi incidents/regulation; key-person risk｜销量与毛利下滑；Robotaxi 事故/监管；关键人物风险
- **证伪条件 Falsifiers**: Robotaxi fleet/cities not expanding by guided dates; auto GM ex-credits declining; storage deployments flat｜Robotaxi 未按期扩张；扣除碳积分汽车毛利下降；储能装机持平

2. 产品、客户与收入模式

- 主题 Theme: EV / Autonomy / Robotics / Energy storage 电动车/自动驾驶/机器人/储能
- 业务分部 Segments: Automotive 汽车；Energy storage 储能；Services & other 服务；FSD/Robotaxi option 自动驾驶期权；Optimus option 人形机器人期权
- 公司描述 (DATA, Yahoo/公司申报): Tesla, Inc. designs, develops, manufactures, leases, and sells electric vehicles, and energy generation and storage systems in the United States, China, and internationally. The company operates in two segments, Automotive; and Energy Generation and Storage. The company offers electric vehicles, as well as sells automotive regulatory credits; and non-warranty maintenance services and collision, automotive insurance services, as well as part sales and retail merchandise sale. It also provides sedans and sport utility vehicles through direct and used vehicle sales, a network of Tesla Superchargers, and in-app upgrades; purchase financing and leasing services; services for electric vehicles through its company-owned service locations and Tesla

mobile service technicians; and vehicle limited warranties and extended service plans. In addition, the company engages in the design, manufacture, i

- 主要客户: Retail auto buyers, utilities/IPPs (Megapack), FSD subscribers｜零售购车者、电力公司(Megapack)、FSD 订阅

3. 行业、竞争、监管与供应链 (ANALYST FRAMEWORK)

- 供应商/上游: Battery cells (Panasonic, LGES, CATL), chips (TSMC/Samsung for AI5), lithium｜电芯(松下、LGES、宁德)、芯片(台积电/三星)、锂
- 竞争者 (含业外): BYD, Geely, Waymo (robotaxi), Zoox, Chinese EV makers; Fluence/Sungrow in storage｜比亚迪、吉利、Waymo、Zoox；储能对手 Fluence/阳光电源
- 瓶颈/监管: Robotaxi regulation (state/NHTSA), FSD safety data, cell supply/tariffs｜Robotaxi 法规(州/NHTSA)、FSD 安全数据、电芯供应与关税
- 关键 KPI: Deliveries, auto GM ex-credits, storage GWh, FSD miles, robotaxi cities｜交付量、扣碳积分毛利、储能 GWh、FSD 里程、Robotaxi 城市数
- 零部件现货/合约价: 本版未接入付费价格源(TrendForce/DRAMeXchange 等)——标注 **UNAVAILABLE**，在 KPI 监控中以定性方式跟踪。

4. 历史财务与现金流 (CONFIRMED, 季度, USD)

季度	收入	毛利	营业利润	净利润	FCF	营业利润率
2026-06-30	28.2 bn USD	4.8 bn USD	0.4 bn USD	1.1 bn USD	-1.1 bn USD	1.4%
2026-03-31	22.4 bn USD	4.7 bn USD	0.9 bn USD	0.5 bn USD	1.4 bn USD	4.2%
2025-12-31	24.9 bn USD	5.0 bn USD	1.6 bn USD	0.8 bn USD	1.4 bn USD	6.3%
2025-09-30	28.1 bn USD	5.1 bn USD	1.9 bn USD	1.4 bn USD	4.0 bn USD	6.6%
2025-06-30	22.5 bn USD	3.9 bn USD	0.9 bn USD	1.2 bn USD	0.1 bn USD	4.1%
2025-03-31	N/A	N/A	N/A	N/A	N/A	N/A

资产负债: 现金 43.5 bn USD；总负债 16.1 bn USD；净现金 27.4 bn USD。TTM FCF 5.8 bn USD（FCF 率 5.6%）。

5. 最新财报: 实际 vs 一致预期 vs 指引

季度	EPS 实际	EPS 一致预期	超预期%
2025-09-30	0.5	0.55885	-0.105299994
2025-12-31	0.5	0.45061	0.1096
2026-03-31	0.41	0.34999	0.1715
2026-06-30	0.33	0.54236	-0.39150003

- 管理层指引 Guidance: 自动数据源不含指引——**UNAVAILABLE（待从公司 IR 新闻稿录入）**。
- 下次财报日 (Yahoo, TENTATIVE 直至公司确认): **2026-10-22**；本季 EPS 一致预期 0.45223（24 位分析师）；收入一致预期 28.1 bn USD。

6. 未来 2/4/6/8 季度驱动与价格合理性

期限	情景	期限价格现值	相对现价
2Q	Bear	$32.06	-91.0%
2Q	Base	$57.91	-83.7%
2Q	Bull	$87.50	-75.3%
4Q	Bear	$31.77	-91.0%
4Q	Base	$60.94	-82.8%
4Q	Bull	$95.60	-73.0%
6Q	Bear	$31.47	-91.1%
6Q	Base	$64.00	-82.0%
6Q	Bull	$104.05	-70.7%
8Q	Bear	$31.16	-91.2%
8Q	Base	$67.10	-81.1%
8Q	Bull	$112.86	-68.2%

现价隐含的 EV/NTM-FCF (Base 路径):	期限
2Q	159.1x
4Q	140.3x
6Q	124.1x
8Q	110.2x

解读: 如果 8 季度后 Base 情景所需倍数仍远高于 25x，说明价格依赖更远期的增长。

7. 三情景 Bear / Base / Bull (HYPOTHESIS)

情景	Y1 增速	长期 FCF 率	WACC	永续 g	概率	每股价值	相对现价
Bear	6.8%	8.0%	14.1%	2.5%	25.0%	$27.58	-92.2%
Base	13.5%	10.0%	13.1%	3.0%	50.0%	$44.55	-87.4%
Bull	17.6%	11.5%	12.6%	3.5%	25.0%	$62.52	-82.4%

概率加权价值 **$44.80**（相对现价 -87.4%）。

8. 估值: DCF / Reverse DCF / ROIC-WACC / 同业 / 敏感度

- DCF Base: EV 148.5 bn USD；终值占比 45.7%。
- **Reverse DCF 反向 DCF**: 在 Base 利润率与 WACC 下，现价隐含 Y1 收入增速 ≈ **78.2%**；一致预期 FY+1 增速 13.5%。
- WACC (CAPM) = rf 5.1% + β 1.60 × ERP 5.0% = 13.1%。
- SOTP segments 分部: Automotive 汽车; Energy storage 储能; Services & other 服务; FSD/Robotaxi option 自动驾驶期权; Optimus option 人形机器人期权. **Status: framework only** — segment revenues are INPUT REQUIRED from the latest 10-K/10-Q segment note (not auto-fed, left blank to avoid fabrication)｜分部收入需按最新财报手动录入，未自动填充以避免编造。
- 同业比较 Comps (DATA): | Ticker | 市值 | EV/Sales | Fwd P/E | 收入增速 | 毛利率 | FCF 收益率 | |—|—|—|—|—|—|—| | TSLA | 1,401.0 bn USD | 13.3x | 163.6x | 25.5% | 18.9% | 0.4% | | BYDDY | 85.0 bn USD | 0.1x | 11.3x | -3.2% | 17.9% | -113.3% | | GM | 73.7 bn USD | 1.0x | 5.6x | 1.9% | 10.2% | 29.8% | | RIVN | 18.0 bn USD | 3.1x | -7.2x | 27.2% | 7.5% | -8.6% | | UBER | 143.7 bn USD | 2.8x | 15.9x | 12.2% | 40.8% | 5.0% |
- 敏感度: 见 Excel `Sensitivity` 表 (WACC × 永续增长率)。

9. 风险、反方论证与证伪

- 反方观点 Counter-thesis: Auto volumes/margins keep falling; robotaxi incidents/regulation; key-person risk｜销量与毛利下滑；Robotaxi 事故/监管；关键人物风险
- 证伪指标: Robotaxi fleet/cities not expanding by guided dates; auto GM ex-credits declining; storage deployments flat｜Robotaxi 未按期扩张；扣除碳积分汽车毛利下降；储能装机持平

- 模型风险: 长期利润率与 WACC 为假设；终值占比高时估值对假设极敏感。

10. Source Ledger 来源账本

Item	Value	Source/time
Price 股价	354.7	Yahoo Finance quote, fetched 2026-10-17T07:55 SGT
Market cap 市值	1.401e+12	Yahoo info.marketCap 2026-10-17T07:55
Cash 现金	4.352e+10	Yahoo info.totalCash (latest filing) 2026-10-17T07:55
Debt 负债	1.608e+10	Yahoo info.totalDebt (latest filing) 2026-10-17T07:55
Revenue TTM 收入	1.036e+11	Yahoo info.totalRevenue (TTM, USD) 2026-10-17T07:55
FCF TTM 自由现金流	5.755e+09	Sum of last 4 quarterly FCF (q_cash); quarters ['2026-06-30', '2026-03-31', '2025-12-31', '2025-09-30'] (USD)
Beta	1.916	Yahoo info.beta (5Y monthly)
Risk-free 10Y 无风险利率	0.051	FRED DGS10 2026-10-15
FX USD per USD	1	n/a (USD)
Consensus rev growth FY0	0.1303	Yahoo revenue_estimate 0y
Consensus rev growth FY+1	0.1352	Yahoo revenue_estimate +1y

11. Assumption Register 假设登记表

Assumption	Value	Rationale
ERP 股权风险溢价	0.05	ASSUMPTION: 5.0% house mature-market ERP
Beta clamp	[0.8,1.6] → 1.60	ASSUMPTION: limit extreme betas

Assumption	Value	Rationale
Tax rate (ROIC)	0.18	ASSUMPTION: blended cash tax 18%
Year-1 growth (Base)	0.1352	Consensus FY+1 revenue growth (Yahoo revenue_estimate '+1y'); capped at 100%
Growth fade	linear Y1→terminal by Y10	ASSUMPTION
Starting FCF margin	0.05554	DATA: FCF TTM / Revenue TTM, floored -50%
Long-run FCF margin (Base)	0.1	ASSUMPTION (analyst framework, reached by Y5; see profile)
Bear/Bull Y1 growth multiplier	0.5x / 1.3x	ASSUMPTION
Bear/Bull margin multiplier	0.8x / 1.15x	ASSUMPTION
WACC adj Bear/Bull	+1.0pp / -0.5pp	ASSUMPTION
Terminal g Bear/Base/Bull	2.5% / 3.0% / 3.5%	ASSUMPTION
Scenario probabilities	25% / 50% / 25%	ASSUMPTION
Horizon exit EV/NTM-FCF Bear/Base/Bull	15x / 25x / 35x	ASSUMPTION
Per-share conversion	value × fx ÷ market cap × price	METHOD: avoids ADR/share-class mismatch

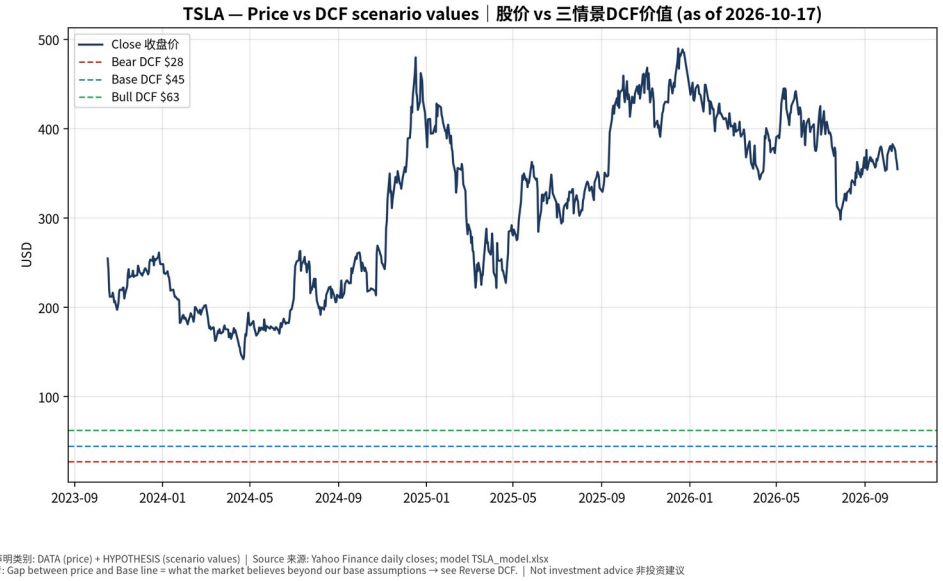
12. Model Provenance 模型溯源

- Model file: `TSLA_model_v20261017-0755.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions)
- Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx)
- Data snapshot: `data/snapshots/20261017-0755` (immutable) | Fetched 2026-10-17 07:55 SGT | Generated 2026-10-17 07:56 SGT
- Version: v20261017-0755. Prior versions are never overwritten (see company history page).

13. 版本与事件时间轴

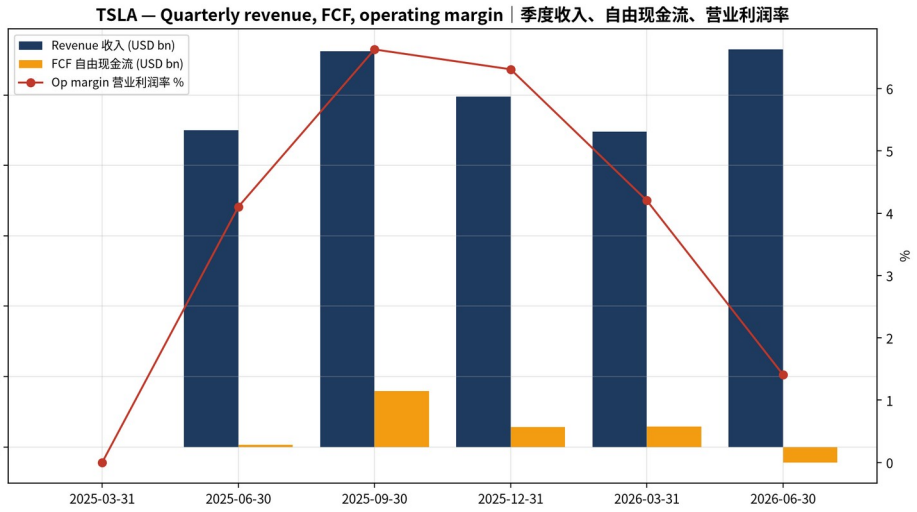
- v20261017-0755: 首版生产系统报告 (27 家同步)。下一事件: 财报 2026-10-22 (TENTATIVE)。

图片报告 Image Report



1_price_vs_value

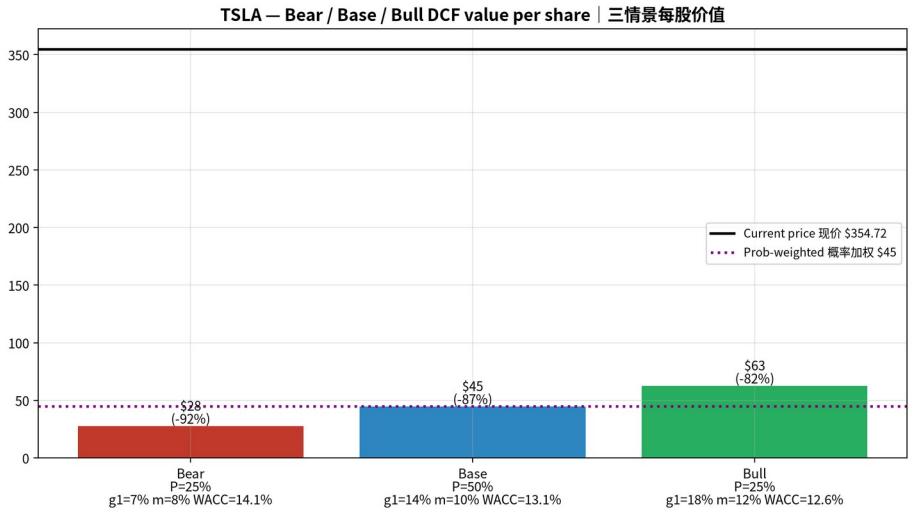
股价vs 三情景价值Price vs scenario values



Claim class 声明类别: CONFIRMED (reported) | Source 来源: Company filings via Yahoo Finance quarterly statements
Thinking 思考: Is FCF growing with revenue? Divergence = capex/working-capital intensity → check cash-flow statement. | Not investment advice 非投资建议

2_financials

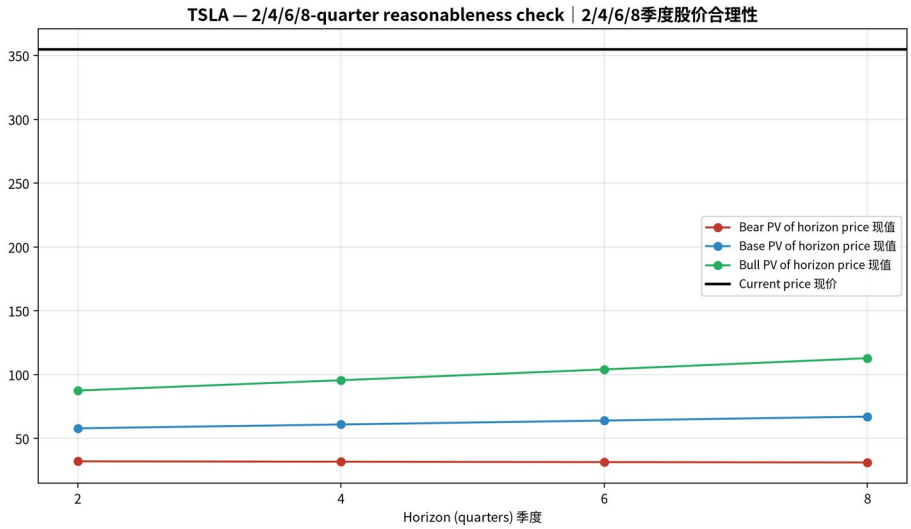
季度财务 Quarterly financials



Claim class 声明类别: HYPOTHESIS (model output) | Source 来源: Model (live formulas) — inputs in Sources/Assumptions sheets
Thinking 思考: Which assumption moves value most? Check Sensitivity sheet; scenario width = uncertainty. | Not investment advice 非投资建议

3_scenarios

三情景 Scenarios



Claim class 声明类别: HYPOTHESIS | Source 来源: Model Horizons sheet; exit EV/NTM-FCF 15x/25x/35x (ASSUMPTION)
Thinking 思考: If even Bull PV < price at 8Q, price requires assumptions beyond Bull within 2 years. | Not investment advice 非投资建议

4_horizons

2/4/6/8 季度检验Horizon check



Reverse DCF 反向DCF: price implies Y1 growth ≈ 78%

Consensus FY+1 growth 一致预期: 14% → gap = expectation the price embeds 价格隐含预期差

Claim class 声明类别: INFERENCE / HYPOTHESIS | Source 来源: Profile (analyst framework) + model
Thinking 思考: Read left→right: each box must have evidence; the falsifier box tells you what would prove the thesis wrong. | Not investment advice 非投资建议

5_logic_map

研究逻辑图Logic map

TSLA Tesla, Inc. — Equity Deep Dive (Professional English) v20261017-0755

Data as of 2026-10-17 07:55 SGT | Price $354.72 | Mkt cap 1,401.0 bn USD | Reporting currency USD | Not investment advice.

1. Investment thesis (HYPOTHESIS)

Core auto margins are compressed by price competition; valuation mostly rests on Robotaxi/FSD and Optimus optionality plus fast-growing storage. The stock is highly reflexive to autonomy milestones.

- **Bull drivers**: Unsupervised FSD scales across cities; Robotaxi unit economics proven; storage margins stay high｜无监督 FSD 多城扩张；Robotaxi 单位经济验证；储能高毛利
- **Bear drivers**: Auto volumes/margins keep falling; robotaxi incidents/regulation; key-person risk｜销量与毛利下滑；Robotaxi 事故/监管；关键人物风险
- **Falsifiers**: Robotaxi fleet/cities not expanding by guided dates; auto GM ex-credits declining; storage deployments flat｜Robotaxi 未按期扩张；扣除碳积分汽车毛利下降；储能装机持平

2. Products, customers, revenue model

- Theme: EV / Autonomy / Robotics / Energy storage 电动车/自动驾驶/机器人/储能 | Segments: Automotive 汽车; Energy storage 储能; Services & other 服务; FSD/Robotaxi option 自动驾驶期权; Optimus option 人形机器人期权
- Business description (DATA): Tesla, Inc. designs, develops, manufactures, leases, and sells electric vehicles, and energy generation and storage systems in the United States, China, and internationally. The company operates in two segments, Automotive; and Energy Generation and Storage. The company offers electric vehicles, as well as sells automotive regulatory credits; and non-warranty maintenance services and collision, automotive insurance services, as well as part sales and retail merchandise sale. It also provides sedans and sport utility vehicles through direct and used vehicle sales, a network of Tesla Superchargers, and in-app upgrades; purchase financing and leasing services; services for electric vehicles through its company-owned service locations and Tesla mobile service technicians; and vehicle limited warranties and extended service plans. In addition, the company engages in the design, manufacture, i
- Customers: Retail auto buyers, utilities/IPPs (Megapack), FSD subscribers｜零售购车者、电力公司(Megapack)、FSD 订阅

3. Industry, competition, regulation, supply chain (ANALYST FRAMEWORK)

- Suppliers: Battery cells (Panasonic, LGES, CATL), chips (TSMC/Samsung for AI5), lithium｜电芯(松下、LGES、宁德)、芯片(台积电/三星)、锂
- Competitors: BYD, Geely, Waymo (robotaxi), Zoox, Chinese EV makers; Fluence/Sungrow in storage｜比亚迪、吉利、Waymo、Zoox；储能对手 Fluence/阳光电源
- Bottlenecks/regulation: Robotaxi regulation (state/NHTSA), FSD safety data, cell supply/tariffs｜Robotaxi 法规(州/NHTSA)、FSD 安全数据、电芯供应与关税
- KPIs: Deliveries, auto GM ex-credits, storage GWh, FSD miles, robotaxi cities｜交付量、扣碳积分毛利、储能 GWh、FSD 里程、Robotaxi 城市数
- Component spot/contract prices: paid feeds not connected — **UNAVAILABLE**, tracked qualitatively.

4. Historical financials & cash flow (CONFIRMED, quarterly, USD)

Quarter	Revenue	Gross profit	Op income	Net income	FCF	Op margin
2026-06-30	28.2 bn USD	4.8 bn USD	0.4 bn USD	1.1 bn USD	-1.1 bn USD	1.4%
2026-03-31	22.4 bn USD	4.7 bn USD	0.9 bn USD	0.5 bn USD	1.4 bn USD	4.2%
2025-12-31	24.9 bn USD	5.0 bn USD	1.6 bn USD	0.8 bn USD	1.4 bn USD	6.3%
2025-09-30	28.1 bn USD	5.1 bn USD	1.9 bn USD	1.4 bn USD	4.0 bn USD	6.6%
2025-06-30	22.5 bn USD	3.9 bn USD	0.9 bn USD	1.2 bn USD	0.1 bn USD	4.1%
2025-03-31	N/A	N/A	N/A	N/A	N/A	N/A

Cash 43.5 bn USD; debt 16.1 bn USD; TTM FCF 5.8 bn USD (FCF margin 5.6%).

5. Latest earnings: actual vs consensus vs guidance

Quarter	EPS actual	EPS consensus	Surprise
2025-09-30	0.5	0.55885	-0.105299994
2025-12-31	0.5	0.45061	0.1096

Quarter	EPS actual	EPS consensus	Surprise
2026-03-31	0.41	0.34999	0.1715
2026-06-30	0.33	0.54236	-0.39150003

- Management guidance: not in automated feed — **UNAVAILABLE (to be entered from IR release)**.
- Next earnings (Yahoo; TENTATIVE until company-confirmed): **2026-10-22**; current-quarter EPS consensus 0.45223 (24 analysts); revenue consensus 28.1 bn USD.

6. 2/4/6/8-quarter drivers & price reasonableness

Horizon	Scenario	PV of horizon price	vs current
2Q	Bear	$32.06	-91.0%
2Q	Base	$57.91	-83.7%
2Q	Bull	$87.50	-75.3%
4Q	Bear	$31.77	-91.0%
4Q	Base	$60.94	-82.8%
4Q	Bull	$95.60	-73.0%
6Q	Bear	$31.47	-91.1%
6Q	Base	$64.00	-82.0%
6Q	Bull	$104.05	-70.7%
8Q	Bear	$31.16	-91.2%
8Q	Base	$67.10	-81.1%
8Q	Bull	$112.86	-68.2%

Current-price implied EV/NTM-FCF on Base path:	Horizon
2Q	159.1x
4Q	140.3x
6Q	124.1x
8Q	110.2x

7. Bear / Base / Bull (HYPOTHESIS)

Scenario	Y1 growth	LR FCF margin	WACC	Terminal g	Prob.	Value/ share	vs price
Bear	6.8%	8.0%	14.1%	2.5%	25.0%	$27.58	-92.2%
Base	13.5%	10.0%	13.1%	3.0%	50.0%	$44.55	-87.4%

Scenario	Y1 growth	LR FCF margin	WACC	Terminal g	Prob.	Value/ share	vs price
Bull	17.6%	11.5%	12.6%	3.5%	25.0%	$62.52	-82.4%

Probability-weighted value **$44.80** (-87.4% vs price).

8. Valuation: DCF / Reverse DCF / ROIC-WACC / comps / sensitivity

- Base DCF EV 148.5 bn USD; terminal share 45.7%.
- **Reverse DCF**: at Base margin/WACC the price implies Y1 revenue growth ≈ **78.2%** vs consensus FY+1 13.5%.
- WACC (CAPM) = rf 5.1% + β 1.60 × ERP 5.0% = 13.1%.
- SOTP segments: Automotive 汽车; Energy storage 储能; Services & other 服务; FSD/Robotaxi option 自动驾驶期权; Optimus option 人形机器人期权 — framework only; segment revenue inputs pending from filings.
- Comps (DATA): see table in Chinese section / Excel `Comps`.
- Sensitivity: Excel `Sensitivity` (WACC × terminal g).

9. Risks, counterarguments, falsifiers

Auto volumes/margins keep falling; robotaxi incidents/regulation; key-person risk｜销量与毛利下滑；Robotaxi 事故/监管；关键人物风险 | Falsifiers: Robotaxi fleet/cities not expanding by guided dates; auto GM ex-credits declining; storage deployments flat｜Robotaxi 未按期扩张；扣除碳积分汽车毛利下降；储能装机持平

10–12. Source Ledger / Assumption Register / Model Provenance

See tables above (shared, bilingual). - Model file: `TSLA_model_v20261017-0755.xlsx` (live formulas; sheets Inputs→Scenarios→DCF→ReverseDCF→Sensitivity→Horizons→SOTP→Comps→ROIC_WACC→History→Sources→Assumptions) - Engine: `production/pipeline/model.py` (Python mirror; values in this report = Python mirror, verified against LibreOffice recalculation of the xlsx) - Data snapshot: `data/snapshots/20261017-0755` (immutable) | Fetched 2026-10-17 07:55 SGT | Generated 2026-10-17 07:56 SGT - Version: v20261017-0755. Prior versions are never overwritten (see company history page).